Rishi Raj, PhD

Theoretical Physics | Quantitative Research | C++/Python, ML/AI, Mathematical Modelling

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Theoretical physics PhD from Sorbonne University with peer-reviewed work on supersymmetric black holes, scientific machine learning, and high-performance C++ simulation. Experienced in translating abstract mathematics into concrete algorithms, validating complex research workflows, and extracting structure from large datasets. Drawn to quantitative research as a natural setting to apply mathematical rigour, engineering discipline, and fast learning to real-world data-driven problems.

EDUCATION

Participant, CERN STEAM Academy 2026 — Jun–Aug 2026 (upcoming)
CERN — Geneva, Switzerland
Selected for highly competitive 10-week CERN doctoral programme in advanced software engineering, HPC/edge computing, and AI/ML, covering C++/Python, MLOps, FPGA systems, transformers, GNNs, anomaly detection, and generative models.

PhD in Theoretical Physics — Oct 2023 – Jun 2026
Sorbonne University / LPTHE / EDPIF — Paris, France
Thesis: *BPS Black Holes and Donaldson–Thomas invariants*. Defended on June 10, 2026.

B.S. (Hons.) & M.S. in Theoretical Physics (Minor in Mathematics) — 2018 – 2023
Indian Institute of Technology, IIT Madras — Chennai, India — M.S. GPA: 9.59/10; Overall GPA: 9.18/10
Master's thesis: *Black Holes through the lens of Matrix Theory*.

SELECTED PROJECTS

Learning Holographic Horizons: Time-Series Modelling, Feature Engineering, and Scientific ML — 2023
with Vishnu Jejjala, Ayan Mukhopadhyay, Sukrut Mondkar; Published in Phys. Rev. D (2025)

- *Physics-informed ML:* generated training data by solving asymptotically AdS_4 Einstein equations in characteristic form (Chesler–Yaffe), with pseudospectral radial discretization (Chebyshev grid) and Adams–Bashforth time stepping; extracted boundary pressure-anisotropy time series and computed apparent/event horizon areas.
- *Modelling + evaluation:* compared long-sequence models against compact feature-based representations, showing that a 10-dimensional engineered representation nearly matched models trained on full time-series inputs; validated performance with strict splits, k-fold cross-validation, and robustness checks.

Black Holes through the lens of Matrix Theory: High-performance C++ simulation — 2022–23
Master's thesis — manuscript

- *C++ simulation suite (BFSS dynamics):* implemented real-time multi-matrix ODE evolution of a strongly-coupled, chaotic many-body system using Boost/scientific libraries; enforced Gauss-law constraints and monitored conserved quantities (energy, angular momentum) as trajectory-level correctness checks across large stochastic ensembles.
- **HPC-scale trajectory generation:** scaled to matrix sizes up to $N \sim 15$ and generated large trajectory ensembles via week-long PBS batch campaigns on the IITM Aqua HPC cluster (64+ cores), enabling statistical characterization of dynamical regimes and thermalization timescales.
- *Performance engineering:* migrated prototypes from Mathematica $\rightarrow$ Python/Julia/C $\rightarrow$ optimized C++, achieving $\sim$100–200$\times$ speedups over the initial baseline while improving numerical stability and verification coverage.

BPS Dendroscopy: Algorithmic reconstruction and consistency validation — 2024
with Boris Pioline, Bruno Le Floch @ LPTHE; Published in Ann. Henri Poincaré (2026)

- *Algorithmic reconstruction:* developed computational methods where local consistency constraints determine global invariants from initial data; constructed and compared solutions across distinct parameter regimes.
- *Verification + tooling:* implemented systematic enumeration/consistency checks (ray-tracing, mutation bookkeeping, cutoffs) and performed broad numerical validation across moduli; developed Mathematica workflow to regenerate figures/tables.

PUBLICATIONS

[*]Equal contribution works; author name appears alphabetically in hep-th

- Vishnu Jejjala, Sukrut Mondkar, Ayan Mukhopadhyay, and Rishi Raj (2025). "Learning holographic horizons". In: *Phys. Rev. D* 111.2, p. 026016. DOI: `10.1103/PhysRevD.111.026016`. arXiv: `2312.08442 [hep-th]`
- Boris Pioline and Rishi Raj (2026). "Black hole quantum mechanics and generalized error functions". In: *JHEP* 03, p. 179. DOI: `10.1007/JHEP03(2026)179`. arXiv: `2507.08551 [hep-th]`
- Bruno Le Floch, Boris Pioline, and Rishi Raj (2026). "BPS Dendroscopy on Local $\mathbb{P}^1 \times \mathbb{P}^1$". In: *Annales Henri Poincaré*. DOI: `10.1007/s00023-026-01666-3`. arXiv: `2412.07680 [hep-th]`

RESEARCH AND WORK EXPERIENCE

Doctoral Researcher Oct 2023 – Jun 2026

Sorbonne University, Laboratory of Theoretical and High Energy Physics (LPTHE) — Paris, France

- *Quantitative modelling:* developed models of high-dimensional systems using advanced geometry, modularity, wall-crossing, and supersymmetric quantum mechanics; translated abstract theory into computable predictions.
- *Algorithm design:* created symbolic and numerical methods to compute scattering diagrams and enumerate BPS spectra, including gradient-flow algorithms for evolving stability-space trajectories from known asymptotic limits.
- *Scientific computing:* implemented reproducible research workflows and libraries in Mathematica for parameter scans, asymptotic expansions, numerical experiments, and cross-validation across independent formulations.
- *Complex data interpretation:* analysed large computational outputs, extracted robust mathematical structure, and distilled results into clear figures, tables, and testable conjectures leading to rigorous proofs.
- *Publication and communication:* co-authored peer-reviewed research papers and presented technical results in conferences, seminars, and journal clubs, combining mathematical depth with clear scientific communication.

Research Intern (Theoretical Physics) May – Jul 2023

Sorbonne University, LPTHE / LPENS — Paris, France

- *Charpak Lab Fellow:* developed algorithms for attractor-flow tree enumeration, modelling multi-centred supergravity black holes as rooted binary trees and resolving subtleties such as scaling solutions and fake transformation walls in a conjectural partition of Denef moduli space.

Teaching Assistant Jul 2022 – Apr 2023

Department of Physics, IIT Madras — Chennai, India

- *Computational physics teaching:* led Python/Mathematica lab tutorials on numerical simulation and data analysis, coaching students on debugging, validation, scientific reporting, and clear technical communication.

SELECTED TRAINING, FELLOWSHIPS AND AWARDS

Training: **4EU+ Ph.D. Summer School on Machine Learning in Physics** (Jun 2025; Niels Bohr Institute, Copenhagen, DK); **1st ROOT Hackathon: The Fixathon** (Feb 2024; CERN, Geneva, CH).

Awards/fellowships: **Charpak Lab Fellowship** (2023; Campus France / French Government); **63rd and 64th Institute Day Prizes** (2023; IIT Madras); **Microsoft Code.Fun.Do++ Hackathon** (2018; Microsoft India / IIT Madras); **KVPY & NIUS Fellowships** (2018–2023; DST, Govt. of India; HBCSE–TIFR).

SKILLS

Core programming: Python for numerical/data workflows; C++17/20 for performance-critical scientific code; Bash, SQL, Git/GitHub, Linux

Scientific Python / ML: NumPy, SciPy, pandas, Matplotlib, scikit-learn, PyTorch; supervised learning, time-series modelling and feature engineering, experiment tracking and metric evaluation

Quantitative methods: stochastic process modelling, statistical hypothesis testing, cross-validation, model diagnostics, parameter-space scans, numerical model validation, linear algebra and optimization, probability theory

Scientific computing: ODE/PDE simulations, lattice/FFT-based numerical methods, symbolic and numerical workflows in Mathematica; ROOT, Boost, GSL

HPC / reproducibility: CUDA, MPI/Open MPI, PBS batch jobs; CMake/Make, Docker, GitHub Actions, pytest, conda/uv/vcpkg, reproducible figure and data pipelines

Communication: peer-reviewed scientific writing, technical presentations, collaborative research; English fluent, German A1, French A1

Last updated: June 12, 2026